

# Handwritten Hindko Text Recognition Using Deep Neural Network



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# Handwritten Hindko Text Recognition using Deep Neural Network

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In partial fulfillment  
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## MS (Artificial Intelligence)

By

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A Post Graduate Thesis submitted to the department of Computer Science and Information Technology as partial fulfillment of the requirement for the award of Degree of MS in Artificial Intelligence.

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## **Declaration**

I Tanveer Ahmed, Registration No. M21F0173AI022, hereby declare that I have produced the work presented in this thesis, during the scheduled period of study. I also declare that I have not taken any material from any source except referred to wherever due to that amount of plagiarism in within an acceptable range. If a violation of HEC rules on research has occurred in this thesis, I shall be liable to punishable action under the plagiarism rules of the HEC.

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## Certificate

It is certified that Tanveer Ahmed with Registration No M21F0173AI022 has carried out all the work related to this thesis under my supervision at the Department of Computer Science and Information Technology, Pak-Austria Fachhochschule: Institute of Applied Sciences and Technology and the work fulfills the requirement for award of MS degree.

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## **DEDICATION**

This thesis is dedicated to my family, whose unwavering support, encouragement, and understanding have been the cornerstone of my academic pursuit. Your belief in my abilities has been a constant source of inspiration, guiding me through every challenge and triumph. I am immensely grateful for your love, sacrifices, and belief in my journey.

I also dedicate this work to my mentors and teachers, whose guidance and expertise have shaped my scholarly endeavors. Your wisdom and encouragement have been invaluable in shaping my perspective and fostering my intellectual growth.

Finally, to all those who have touched my life in various ways, directly or indirectly contributing to my academic and personal development, your impact has not gone unnoticed. This thesis stands as a tribute to the collective support and influence that has shaped my educational voyage.

With deepest gratitude,

Tanveer Ahmed

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**Tanveer Ahmed**

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## ABSTRACT

### Handwritten Hindko Text Recognition using Deep Neural Network

Handwritten words recognition is a challenging task with significant implications for cultural preservation and technological advancement. In this research paper, we propose a unique approach for Hindko Handwritten words recognition. The primary objectives of the study were to develop an accurate and efficient system capable of recognizing Hindko handwritten text and to compare the performance of deep learning models, Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) and HTR-Attention model. To achieve these objectives, we collected a substantial dataset of handwritten Hindko words, comprising diverse writing styles and variations. Processing techniques were applied to enhance the data quality and extract meaningful features for training the deep neural networks. Through rigorous experimentation and evaluation, the research findings demonstrated promising results. The HTR-Attention model achieved an impressive accuracy of 88.97%. while CNN and RNN models attained an accuracy of 88.36% and 80.45% respectively. The potential applications of this research encompass various fields, including language recognition systems, optical character recognition systems, and language translation technologies. Furthermore, the developed model can be integrated into digital learning platforms to facilitate Hindko language learners and preserve the cultural heritage associated with this regional Language.

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## LIST OF ABBREVIATIONS

|     |                               |
|-----|-------------------------------|
| OCR | Optical Character Recognition |
| CNN | Convolutional Neural Networks |
| HTR | Handwritten Text Recognition  |
| RNN | Recurrent Neural Networks     |

# **Chapter 1**

## **Introduction**

## 1. Introduction

Handwritten text recognition is the detection of text from images, documents and from other sources and change them in machine-readable form for further processing. Hindko is spoken by more than 5 million people as their first language, according to the 2017 census. It's mostly spoken in Punjab and Khyber Pakhtunkhwa. The written content in Hindko covers a wide range of subjects, including religion, education, poetry, politics, and many more. Hindko still needs a fully developed OCR system instead of all of these.

Although there are a lot of spoken examples of the Northern Pakistani language Hindko, there aren't many written sources available. To protect and promote the heritage and cultural importance of this regional language, creating a useful system for recognizing handwritten Hindko words becomes an important task. Different people write Hindko words in a variety of styles and variations, which makes it hard to make a system that can recognize them correctly and reliably.

The main goal of the study is to find out how deep neural networks like CNN, RNN and HTR Attention model can interpret Handwritten Hindko text. We can extract complex patterns and subtleties from handwritten text that lead to more dependable recognition results, by applying deep neural network models to their fullest capabilities. We can concentrate on CNN (Convolutional Neural Network), HTR Attention Model, and RNN (Recurrent Neural Network) as our main models because they have demonstrated efficacy in range of visual and sequence data processing tasks.

As application of CNNs in Handwritten text recognition makes good sense as they are highly good at extracting spatial characteristics from photographs. Handwritten Hindko character exhibits unique visual patterns and structures and CNNs are very good in extracting them, so CNN is good choice for character-level recognition tasks. Furthermore, CNN has the capability to distinguish between various characters due to their ability to capture hierarchical representation, despite differences in writing styles.

HTR-Attention models dynamically focus on the relevant parts of the input sequence so based on their ability to dynamically focus on relevant part of the input sequence they can be used in Hindko Handwritten text recognition. Character level recognition tasks derive significant advantages from the unique visual attributes and the structure of the Hindko characters, which are well-captured by HTR-Attention models. For further improving the recognition accuracy,

specially in the condition when the handwritten script consists of complicated contextual information or multiple writing styles, HTR-Attention model uses attention technique to allow them dynamically focus on different parts of the data.

RNN is suitable for processing sequential data and it is preferred option for the identification of text that is composed of many characters. Hindko language script is written from right to left and for successfully interpretation of meaning of text it is essential to understand the temporal relationship between characters. These connections can be accurately shown by recurrent neural networks and are anticipated to demonstrate exceptional proficiency in capturing the unique sequential patterns that are observed in Hindko text.

For training and testing of these deep neural networks models a huge collection of Handwritten text is used that consists of variety of writing styles, character shapes, and levels of handwriting clarity. Before feeding the data into the models, preprocessing techniques are employed to standardize the images and extract relevant features to aid in the learning process.

Natural language processing is a field that is actively researching handwritten text recognition, which has many commercial and professional uses. Processing forms at government, administrative, medical, and academic institutions, as well as processing bank checks and recognizing postal addresses, are a few examples of these uses. The process of automatically translating a source language into its symbolic representation is known as handwritten text recognition.

We have long-term research plan, which will lead to a complete OCR system for Hindko. As there's still no standard dataset exists for Hindko OCR development, so as a starting point we will create the first Hindko text dataset. We will use deep learning algorithms for recognition of this text.

### **1.1 Problem Statement:**

Hindko, a language spoken by over 8 million people, lacks the essential technological infrastructure for Optical Character Recognition (OCR) systems, particularly for handwritten text. There haven't been any dedicated efforts to make OCR tools for Hindko, even though there are a lot of speakers. This is holding back the language's progress in many ways. This thesis tries to fill in this important gap and make a big difference in the areas of language documentation, linguistic stability, and the general availability of information in Hindko.

1. **Lack of OCR Development:** In the absence of OCR software for Hindko, digitizing handwritten texts is difficult. The language's integration into the digital realm is thereby impeded. Notwithstanding the substantial speaker population, individuals who employ a particular language lack the most recent OCR technology to facilitate efficient text processing and analysis.
2. **Language Documentation:** To ensure that the language and culture of Hindko are preserved for future generations, the proposed work is crucial. It is more challenging to fully record the language when optical character recognition (OCR) tools are not available, since digitizing historical documents, manuscripts, and other handwritten materials becomes more difficult.
3. **Language Stability:** Handwritten texts are difficult to analyze, translate, and interpret on Hindko without OCR tools. This destabilizes language. OCR will improve language stability by letting people read and understand handwritten texts.
4. **Facilitating Native Speakers:** A computerized optical character reader (OCR) will help people who speak Hindko read, understand, and learn new things written in their language. The idea of linguistic inclusivity is supported by this new technology, which makes it easier for speakers to interact with their written heritage.
5. **Research Contribution to OCR:** The primary objective of this thesis is to offer a valuable contribution to the area of optical character recognition (OCR) by developing a tailored system for Hindko. This will demonstrate the adaptability and practicality of deep neural networks for handling a seldom-used language's handwritten text.
6. **Lack of Research Effort in Hindko Language:** Millions of people speak Hindko, but not nearly enough research is being done on it, especially when it comes to making OCR systems. In order to close this gap, this thesis will begin a full study on Hindko OCR. This will make way for more research in the future.
7. **Absence of Dataset:** The lack of a dedicated dataset is another obstacle that has been encountered during the development of Hindko OCR. The absence of a standardized dataset hampers the training and evaluation of OCR models tailored to Hindko, emphasizing the need for comprehensive data collection and curation.

## 1.2 Research Questions:

The research presents the following research questions:

- 1) How can deep neural networks, specifically CNNs, effectively capture and learn spatial features from handwritten Hindko characters to achieve accurate character-level recognition?
- 2) What is the efficacy of HTR-Attention models in dynamically focusing on relevant parts of handwritten Hindko words, and how does this contribute to improved character-level recognition, especially in cases of varying writing styles and contextual complexities within the script?
- 3) To what extent can RNNs, with their proficiency in handling sequential data, effectively model the temporal dependencies between characters in Hindko words, considering the language's right-to-left writing direction?

### **1.3 Research Contribution:**

The main contribution of this research is in the development and exploration of a specialized Handwritten Text Recognition (HTR) system for the Hindko language, utilizing deep neural network models. The distinctive features and contributions of this study are outlined below:

#### **1.3.1 Novel Application of Deep Neural Networks:**

This research contributes to the field by applying deep neural network models, specifically Convolutional Neural Networks (CNNs), HTR-Attention models and Recurrent Neural Networks (RNNs) to the domain of Hindko handwritten text recognition. The novel application of these models aims to harness the power of deep learning for deciphering the intricate patterns and variations present in handwritten Hindko words.

#### **1.3.2 Customized Model Selection for Hindko Recognition:**

The study carefully chooses and uses CNNs, RNNs, and HTR-Attention models based on how well they can handle the unique problems that Hindko characters and handwriting styles present. Based on the research these models provide sufficient adaptability to be applied in several scenarios and excel in capturing temporal interdependencies, geographical characteristics, and significant contextual intricacies. Hindko OCR technology is enhanced by this.

#### **1.3.3 Creation of a Hindko Handwritten Words Dataset:**

Significant achievement of this research is the development of the Hindko Handwritten words

dataset. As there is no standardized dataset of Hindko language is available so in order to build OCR for Hindko, it is necessary to include a variety of writing styles, character forms, and varying levels of handwriting clarity in this particular dataset. The presence of this dataset will help the future researcher and provide assistance in training and assessment of OCR models.

#### **1.3.4 Addressing Hindko's Technological Gap:**

More than 8 million people speak Hindko language but still Hindko doesn't have any Optical Character Recognition (OCR) tool, so this study will fill a big gap in technology. This study aims to close the gap by developing unique OCR system. Consequently, the conversion of handwritten Hindko text into digital text will play a vital role in modernizing Hindko.

#### **1.3.5 Language Preservation and Documentation:**

The proposed research signifies a notable advancement in the preservation of the Hindko language and guarantees of its documentation. The OCR system for Hindko will help in the process of digitizing historical documents, manuscripts, and various other forms of written materials and this will also help in preservation of Hindko's cultural and linguistic history.

#### **1.3.6 Enhancing Linguistic Stability:**

As people read and understand handwritten documents so the development of an OCR system for Hindko language will help in maintaining the stability of the language. This OCR system will enhance linguistic stability and people can easily read, translate and understand handwritten Hindko documents.

#### **1.3.7 Empowering Native Speakers:**

The main aim of the research is to increase the empowerment of native Hindko speakers by creating a technological tool that can help in their reading, understanding, and acquisition of novel written material in their language. This aligns with the concept of linguistic inclusivity and facilitates interaction between native speakers and their written heritage.

#### **1.3.8 Initiating Comprehensive Research Efforts in Hindko OCR:**

The study initiates a comprehensive investigation to compensate for the insufficient amount of dedicated research conducted by Hindko OCR. This study elucidates the deficiencies in existing research and proposes potential avenues for addressing them. This sets the stage for future research endeavors aimed at developing optical character recognition (OCR) systems for

less commonly utilized languages such as Hindko.

## **Chapter 2**

### **Literature Review**

## **2. Literature Review:**

A lot of work has been done on optical character recognition, but no work has been done on OCR for the Hindko language. The following are some of the things that have been done on OCR of other languages:

The paper<sup>1</sup> describes the PHND dataset, which is a set of handwritten Pashto numbers, along with a deep learning benchmark that is used to test how well the recognition works. The dataset has Pashtu numbers written by different people and marked up by hand for training and testing. For identification jobs, deep learning models like CNNs and RNNs are utilized. The dataset has 50,000 scanned pictures in nine different categories. The tests showed that the recognition was very good; CNN models got as high as 99.21% accuracy on the PHND dataset.

The paper<sup>1</sup> talks about the PHCD dataset, which is a set of handwritten Latin small and capital letters, numbers, and symbols. It uses a Convolutional Neural Network (CNN) as its main tool to solve the recognition problem. There are 530,000 scanned images in the dataset, arranged in groups from A to Z, a to z, and 0 to 9. Through extensive experimentation and evaluation, the researchers achieve promising results and gained 96% of accuracy in recognizing handwritten Latin characters with diacritics using the CNN model.

The paper<sup>3</sup>, focuses on the development of a methodology for recognizing handwritten Urdu characters using Convolutional Neural Networks (CNNs) and transfer learning. The primary goal of the paper is to address the challenge of recognizing handwritten Urdu characters through the method of deep learning techniques, specifically CNNs. The dataset consists of 70120 Samples. The experimental analysis demonstrates that OCR-GoogleNet and OCR-AlexNet, and achieve substantial performance improvements, with average recognition rates of 94.7%, and 96.3% respectively.

The paper<sup>4</sup> presents a novel approach to recognizing handwritten Turkish letters using Convolutional Neural Networks (CNNs). The main goal of the paper is to address the task of recognizing handwritten Turkish letters, a specific character recognition problem, by developing a new CNN-based model. Turkish letter dataset consists of 25875 samples. Three models named VGG19, InceptionV3, and Xception were used. Model achived 96.68% of accuracy.

In the paper<sup>5</sup> referenced as an effective CNN architecture for recognizing handwritten Vietnamese characters is introduced. The model comprises three convolutional layers and two

fully connected layers. To mitigate the risk of overfitting, a dropout technique is integrated into the fully connected layers. The experimentation employs the NIST dataset along with self-collected Vietnamese characters, totaling 72,500 samples. The experimental outcomes demonstrate that the model attained an outstanding accuracy level of 97.2%.

Paper<sup>6</sup> centers its attention on recognition of handwritten Devanagari characters, employing a fine-tuned deep Convolutional Neural Network (CNN). A novel Devanagari character dataset, comprising 5800 images encompassing vowels and consonants, is utilized. The VGG16 architecture of the DCNN is leveraged to automatically detect and categorize key attributes. Remarkably, the fine-tuned VGG16 architecture achieves an outstanding classification accuracy of 96.55%.

**Table 1:** Summary of Related Work

| Sr.No | Paper Title                                                                                            | Dataset/Samples           | Methodology/Model                   | Accuracy |
|-------|--------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------|----------|
| 1     | PHND: Pashtu Handwritten Numerals Database and deep learning benchmark                                 | PHND/50,000               | CNN,RNN, ResNet, GoogleNet, AlexNet | 98%      |
| 2     | Recognition of handwritten Latin characters with diacritics using CNN                                  | PHCD/530,000              | CNN                                 | 96%      |
| 3     | OCR-nets: Variants of Pre-trained CNN for Urdu Handwritten Character Recognition via Transfer Learning | IFHCDB/70120              | OCR-AlexNet, OCR-GoogleNet          | 96.3%    |
| 4     | A novel handwritten Turkish letter recognition model based on convolutional neural network             | TLD/25875                 | VGG19, InceptionV3, and Xception    | 96.68%   |
| 5     | Vietnamese handwritten character recognition using                                                     | NIST dataset + Vietnamese | CNN Architecture                    | 97.2%    |

|   |                                                                                                                                |                                                         |            |        |
|---|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|------------|--------|
|   | convolutional neural network                                                                                                   | Characters/72,500                                       |            |        |
| 6 | Devanagari Handwritten<br>Character Recognition using<br>fine-tuned Deep<br>Convolutional Neural<br>Network on trivial dataset | Handwritten<br>Devanagari<br>Character<br>Dataset/ 5800 | CNN, VGG16 | 96.55% |

## **Chapter 3**

### **Research Methodology**

### **3.0 Research Methodology:**

To create a precise Hindko handwritten character recognition system using deep neural networks, we carefully planned and executed a thorough research methodology. A number of important steps make up the method we use to make sure that the data is reliable and the model works well.

#### **3.1 Data Collection:**

We have collected dataset of Handwritten Hindko text dataset for the purpose of our study, “Hindko handwritten text recognition using deep neural networks”. Twenty different Hindko words that are written by more than thousand of people were collected to make sure the robustness and representativeness of our dataset. For the collection of dataset samples, were taken from the students and instructors from different universities and colleges. The purpose of large collection of datasets was to record the differences in handwriting characteristics like writing styles and stroke widths etc.

#### **3.2 Data Augmentation:**

The enhancing the diversity and size of the dataset various augmentation techniques, such as rotation, width shift, height shift, shear, zoom, and fill mode, were employed. These Augmentation techniques that are used in our dataset will enhance its robustness and expanded its capacity to accommodate a broader spectrum of writing styles and conditions. These methods have effectively introduced nuanced variations into the handwritten characters. Our dataset consisted of a total of 56,657 images following the application of these techniques.

#### **3.3 Data Balancing:**

So that we could be sure the dataset would be fair, the data were split into twenty different groups, and each group had a different Hindko number. There were no class imbalances in the dataset, which is important because they could have led to our deep neural networks being trained incorrectly. Between 2500 and 3000 samples were in each class.

#### **3.4 Data Preprocessing:**

Preprocessing steps were conducted prior to the input of data into our deep learning models. Firstly, we implemented a consistent resizing of all the images in the dataset to a size of 50x50

pixels. As a result of this standardization, subsequent processing steps were streamlined and input dimensions were consistently maintained across models.

### **3.5 Data Conversion to CSV:**

We turned the whole set of images into a CSV (Comma-Separated Values) file so that our deep neural networks could read them. The 2500 pixels in each row of the CSV file were flattened into columns to make a single image. An extra column was also added to show the label or class of each row and the Hindko number that goes with it.

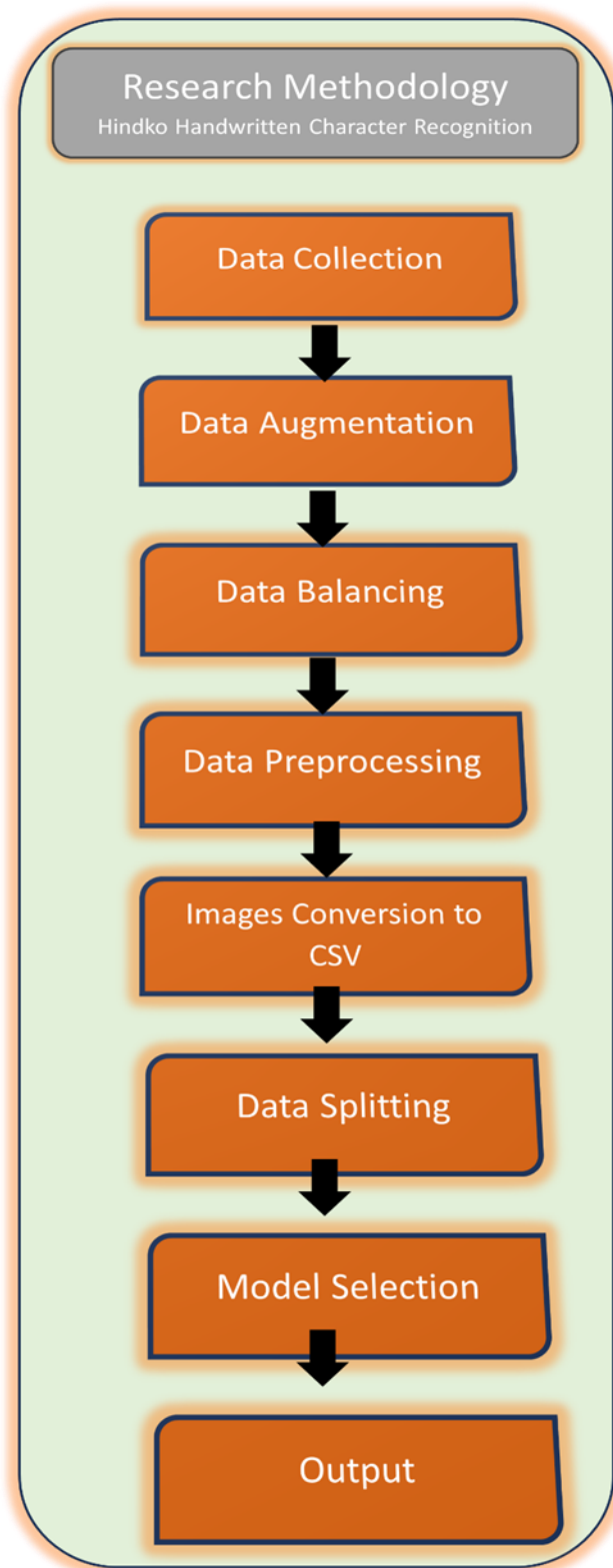


Figure 1: *Research Methodology*

### **3.6 Data Splitting:**

The dataset was split into a training set and a testing set to make testing and training the model easier. For training purpose, we have separated 70 percent of data and for testing we have separated 30 percent of data. This train and test separation of the data will reduce the risk of overfitting.

### **3.7 Model Selection:**

Selection of best model is the important step of every research. In our study we have selected three models that are HTR-Attention Model, CNN Model and RNN model. These models are selected for investigating different methods for identifying Handwritten Hindko text and evaluate their performance on the dataset.

The main purpose of the study is the development of a handwritten Hindko text recognition system that will be able to accurately classify handwritten Hindko text. This aim will be achieved by using complete research methodology.

### **3.8 CNN Model:**

The Convolutional Neural Network (CNN) is a deep neural network that is primarily used for visual tasks such as object recognition, image segmentation, and task labeling. Convolutional Neural Networks (CNNs) have emerged as a notable breakthrough in the field of natural language processing. Convolutional Neural Networks (CNNs) are a type of machine learning algorithm designed specifically for the task of identifying handwritten characters in the Hindko script. In this analysis, the model architecture will be thoroughly examined and a mathematical representation for each layer will be provided.

#### **3.8.1 Convolutional Layer 1:**

The Rectified Linear Unit (ReLU) activation function consists of 32 filters and is utilized in the Convolutional Neural Network (CNN) layer 1. The dimensions of each filter is (3, 3), and input data with dimensions of 50x50 pixels and a single channel is utilized by the procedure.

#### **3.8.2 Mathematical Formulation:**

According to the framework of convolutional layers, let  $W^{[1]}$  denote the weights thereof. The biases of the initial convolutional layer are denoted by  $b^{[1]}$ . Computed as follows is the output of this layer:

$$Z^{[1]} = \text{Conv2D}(X, W^{[1]}) + b^{[1]}$$

$$A^{[1]} = \text{ReLU}(Z^{[1]})$$

### **3.8.3 MaxPooling Layer 1:**

Pooling operations with a size of (2, 2) pick the largest value in each 2x2 window to reduce the spatial dimensions of the layer below it. This is called "downsampling" the feature maps. This method helps pull out the most important features while making the computations simpler..

### **3.8.4 Dropout Layer 1:**

When the dropout rate is 0.25, it means that during training, about 25% of randomly chosen neurons from the layer below are turned off or set to zero at each step. To avoid overfitting and improve model generalization, this dropout technique is used.

### **3.8.5 Convolutional Layer 2:**

In the second convolutional layer, the activation function is ReLU and 64 filters with a size of (3, 3) are utilized. Using the output from the previous layer as input, this layer detects higher-level features in the data, much like Convolutional Layer 1.

### **3.8.7 MaxPooling Layer 2:**

The second layer of MaxPooling uses the same pooling size as the first layer, which is (2, 2). By picking the highest value in each 2x2 window, it helps with feature abstraction and dimensional reduction by reducing the feature maps' spatial dimensions..

### **3.8.8 Dropout Layer 2:**

Similar to Dropout Layer 1, Dropout Layer 2 mimics its behavior by using a dropout rate of 0.25. As a precaution against overfitting, it trains with about 25% of the neurons from the previous layer deactivated at random..

### **3.8.9 Convolutional Layer 3:**

Convolutional Layer 3 employs 128 filters of size (3, 3) with ReLU activation, resembling the configuration of Convolutional Layer 2. It continues to extract higher-level features from the previous layer's output.

### 3.8.10 MaxPooling Layer 3:

MaxPooling Layer 3 employs a (2, 2) pooling size, similar to MaxPooling Layer 1, effectively reducing the spatial dimensions of the feature maps by selecting the maximum value in each 2x2 window, aiding in feature abstraction and dimension reduction.

### 3.8.11 Dropout Layer 3

Dropout Layer 3 utilizes a dropout rate of 0.25, following the same pattern as Dropout Layer 1. During training, it randomly deactivates around 25% of the neurons from the preceding layer to prevent overfitting and promote model generalization.

### 3.8.12 Flatten Layer:

In order to get ready for fully connected layers, the flatten layer flattens the output from the preceding layer into a 1D vector.

### 3.8.13 Dense Layer 1:

Dense Layer 1 consists of 128 neurons with ReLU activation. Following is the Mathematical Formulation of Dense Layer 1:

$$Z^{[2]} = W^{[2]} \cdot A^{[10]} + b^{[2]}$$
$$A^{[2]} = \text{ReLU}(Z^{[2]})$$

Here  $W^{[2]}$  represent the weights of the first dense layer, and  $b^{[2]}$  represent the biases of the first dense layer.

### 3.8.14 Dropout Layer 4:

Dropout Layer 4 utilizes a dropout rate of 0.5, following the same behavior as Dropout Layer 1. During training, it randomly deactivates half of the neurons from the preceding layer to prevent overfitting and enhance the model's generalization ability.

### 3.8.15 Dense Layer 2:

Dense Layer 2 consists of 20 neurons with a softmax activation function, typically used for classification tasks to produce probability distributions over multiple classes. Following is the mathematical Formulation in which  $W^{[3]}$  represent the weights of the second dense layer, and  $b^{[3]}$  represent the biases of the second dense layer.

$$Z^{[3]} = W^{[3]} \cdot A^{[2]} + b^{[3]}$$

$$A^{[3]} = \text{Softmax}(Z^{[3]})$$

The sparse categorical cross-entropy loss function and Adam optimizer are used to compile the model after the architecture has been defined. The training data (X\_train and y\_train) provided are then used to train it for 50 epochs with a batch size of 400, and the validation data (X\_valid and y\_valid) are used to validate it. The training history is stored in the 'history' variable, which allows one to display validation and training performance across epochs..

### **3.9 HTR-Attention Model:**

The HTR-Attention (Handwritten Text Recognition with Attention) model is a deep learning system created specifically to identify handwritten text. Our model incorporates a one-of-a-kind attention mechanism with convolutional neural networks (CNNs) for feature extraction. This allows us to concentrate on aspects of the input image that are relevant to the prediction process. In this section, we will provide a description of the model as well as its equivalent mathematical formulation.

#### **3.9.1 Model Description:**

The HTR-Attention model can be divided into three main parts:

##### **3.9.2 Convolutional Neural Network (CNN):**

The model takes grayscale images of size 50x50 pixels as input. The input is passed through three convolutional layers with 32, 64, and 128 filters respectively, using ReLU activation functions. After each convolutional layer, max pooling is applied to downsample the feature maps. Dropout layers are used for regularization to prevent overfitting.

##### **3.9.3 Fully Connected Layers:**

The output of the last convolutional layer is flattened to create a one-dimensional vector. There are two dense (completely linked) layers employed, the first having 20 units with softmax activation and 128 units with ReLU activation. The final output probabilities for each of the 20 classes are generated by the softmax layer.

##### **3.9.4 Attention Mechanism:**

Reshape: The output of the first dense layer is reshaped into a one-dimensional vector with a shape of (128,). A dense layer with softmax activation, named 'attention\_probs,' is applied to

the reshaped vector to produce attention weights. These weights represent the importance of different parts of the feature vector. The attention mechanism multiplies the output of the first dense layer with the attention weights element-wise. This step highlights the relevant features and suppresses irrelevant ones.

### 3.9.5 Final Model:

The ultimate model is formed by linking the input to the output of the second dense layer (softmax layer), circumventing the attention mechanism.

### 3.9.6 Training:

Since it involves a multi-class classification challenge, the model is constructed utilizing the Adam optimizer and sparse categorical cross-entropy loss. The training process involves using a training dataset ( $X_{\text{train}}$ ,  $y_{\text{train}}$ ) for 50 epochs with a batch size of 200, and validation is conducted on a distinct dataset ( $X_{\text{test}}$ ,  $y_{\text{test}}$ ).

### 3.9.7 Mathematical Formulation:

#### a) Convolutional Layers:

- Let **Conv1**, **Conv2**, and **Conv3** denote the outputs of the three convolutional layers.
- **Conv1** = Conv2D(Input, 32 filters, (3, 3), ReLU activation)
- **Conv2** = Conv2D(Conv1, 64 filters, (3, 3), ReLU activation)
- **Conv3** = Conv2D(Conv2, 128 filters, (3, 3), ReLU activation)

#### b) Fully Connected Layers:

- Let **Dense1** and **Dense2** denote the outputs of the two dense layers.
- **Flatten** = Flatten(Conv3)
- **Dense1** = Dense(Flatten, 128 units, ReLU activation)
- **Dense2** = Dense(Dense1, 21 units, Softmax activation)

#### c) Attention Mechanism: //

- **attention\_input** = Reshape(Dense1, target\_shape=(128,))
- **attention\_probs** = Dense(attention\_input, 128 units, Softmax activation)
- **attention\_mul** = Multiply([Dense1, attention\_probs])

#### d) Final Model:

- **model** = Model(inputs=Input, outputs=Dense2)

**e) Training:**

- `model.compile(optimizer='adam',loss='sparse_categorical_crossentropy',  
metrics=['accuracy'])`
- `history = model.fit(X_train, y_train, epochs=50, batch_size=200,  
validation_data=(X_test, y_test))`

### **3.11 RNN Model:**

There are several layers in this recurrent neural network (RNN) that were made to do a certain job. These layers include an LSTM layer, a dropout layer, and a dense layer with softmax activation. It helps us sort things into 20 groups for a classification task. It is made up of layers that are stacked on top of each other and run in a straight line. It follows the same order that the layers were added, which in this case is one layer after the other..

#### **3.11.1 Mathematical Formulation of RNN Model**

The given mathematical equations explain the operations and transformations that these layers do during the training and inference processes.

#### **3.11.2 LSTM Layer:**

A recurrent layer called Long Short-Term Memory (LSTM) is used to find data that needs to come in a certain order. It usually works best with sequence data, like text or time series. It takes inputs that are (50, 50), which means that each time step must have 50 features and be 50 bits long. The LSTM layer's most important equations are these:

- **Input Gate ( $i_t$ ):**

$$i_t = \text{sigmoid}(W_i * [h_{(t-1)}, x_t] + b_i)$$

- **Forget Gate ( $f_t$ ):**

$$f_t = \text{sigmoid}(W_f * [h_{(t-1)}, x_t] + b_f)$$

- **Output Gate ( $o_t$ ):**

$$o_t = \text{sigmoid}(W_o * [h_{(t-1)}, x_t] + b_o)$$

- **Candidate Cell State ( $\tilde{C}_t$ ):**

$$\tilde{C}_t = \tanh(W_c * [h_{(t-1)}, x_t] + b_c)$$

- **Cell State ( $C_t$ ):**

$$C_t = f_t * C_{(t-1)} + i_t * \tilde{C}_t$$

- **Hidden State ( $h_t$ ):**

$$h_t = o_t * \tanh(C_t)$$

Input, forget, and output gate activations are shown as  $i_t$ ,  $f_t$ , and  $o_t$ , respectively. At time step  $t$ , the input is  $x_t$ , the hidden state is  $h_t$ , the cell state is  $C_t$ , and the candidate cell state is  $\tilde{C}_t$ .

### 3.11.3 Dropout:

Mathematically speaking, the Dropout layer performs a masking operation during training by arbitrarily setting a percentage of input units to zero. During training, each element of the input is set to zero with probability  $p$ , where  $p$  is the dropout rate (0.5), and scaled by  $1/(1 - p)$  to ensure that the expected sum remains the same.

### 3.11.4 Dense Layer:

A fully linked neural network layer is represented by the Dense layer. In the classification task, each of its 20 output units represents a class. Softmax is the selected activation function, which converts the model's raw output scores into probabilities for each class. The mathematical equation for the softmax function for class  $c$  is:

$$P(y=c) = \exp(\text{score}_c) / \sum(\exp(\text{score}_i)) \text{ for all classes } i$$

### 3.11.5 Compilation:

Compilation involves setting up the training process. The 'adam' optimizer uses the following update rule (Adam optimization algorithm):

- $\theta_{t+1} = \theta_t - \eta * (m_t / (\sqrt{v_t} + \epsilon))$

where  $\epsilon$  is a tiny constant to prevent division by zero,  $m_t$  is the first moment estimate,  $v_t$  is the second moment estimate, and  $\theta_t$  is the model's parameter at time step  $t$ .  $\eta$  represents the learning rate. The cross-entropy loss between the true labels and the predicted probability is determined by the 'sparse\_categorical\_crossentropy' loss function. For a single example, the cross-entropy loss mathematical formula is:  $L(y, p) = -\sum(y_i * \log(p_i))$  for all classes  $i$

Where  $y_i$  is the true label for class  $i$ , and  $p_i$  is the predicted probability for class  $i$ .

### **3.12 Training:**

The model is trained for 50 epochs with a batch size of 20 using the `model.fit` function on a training dataset (`X_train`, `y_train`). The model learns to minimize the specified loss function during training, and its progress is tracked in the history object for additional analysis at a later time.

## **Chapter 4**

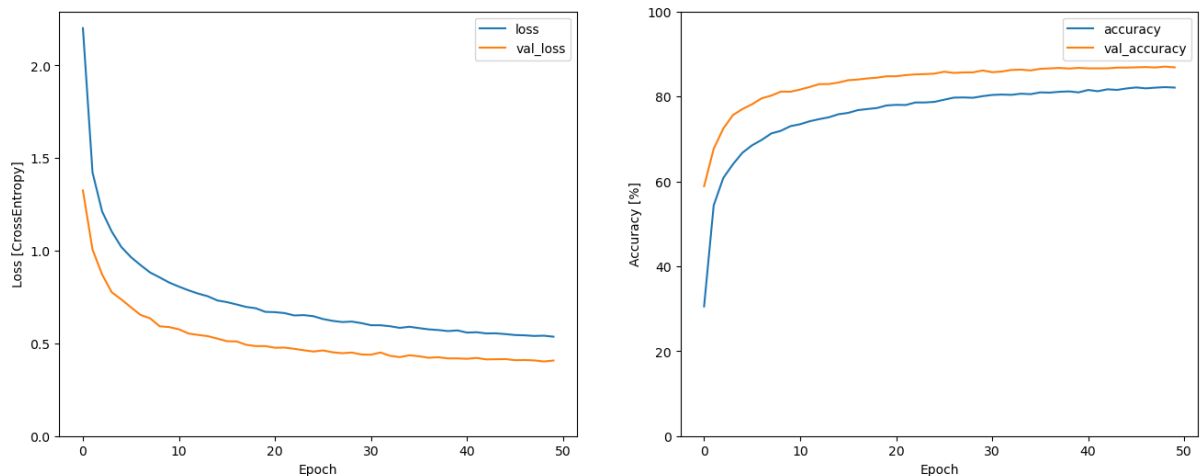
### **Result and Discussions**

## 4.0 Result and Discussions

The findings and conclusions derived from the data for the study "Hindko handwritten character recognition using deep neural networks" are presented in this document. Data collection, augmentation, balancing, preprocessing, and model selection constituted our methodology, as previously described. The valuable insights into the model's performance during training are presented in the two graphs that you have generated using the training history of our Handwritten Text Recognition (HTR) attention model.

### 4.1 Graph 1: Epoch vs Loss

The blue line in the graph represents the training loss over the epochs, indicating the ability of our model to acquire new skills from the training set. The initially large loss indicates that the model's predictions are not very close to the actual values. The fact that the model lost shows that it is getting better at finding patterns in the data and making accurate predictions as it learns. The orange line on the validation dataset serves as an indicator. The validation dataset contains elements that the model has not previously encountered or processed. Given that the model is encountering this data for the first time, it is expected that the validation loss will initially exceed the training loss. However, the validation loss does exhibit a decrease over time, suggesting that the model is proficient in applying to new data that has not yet been analyzed. The model is capable of applying its existing knowledge to new data, as evidenced by the diminishing validation loss. Consequently, it is not solely recollecting the training data.



**Figure 2: Epoch vs Accuracy**

## 4.2 Graph 2: Epoch vs Accuracy

The training accuracy is shown by the blue line graph. It shows the percentage of correctly predicted samples in the training dataset. Just like the training loss, the accuracy starts out lower and then goes up as the model learns. A positive correlation suggests that there is an enhancement in performance when applied to the training dataset. Orange Line represents the accuracy on the validation dataset. Like the validation loss, the validation accuracy is initially lower than the training accuracy but should also show improvement over time. The alignment of the blue and orange lines in this graph suggests that our model is not only learning well on the training data but is also capable of generalizing to new, unseen data. The increasing validation accuracy indicates that the model is performing well on data it hasn't been explicitly trained on.

## 4.3 Models Accuracy:

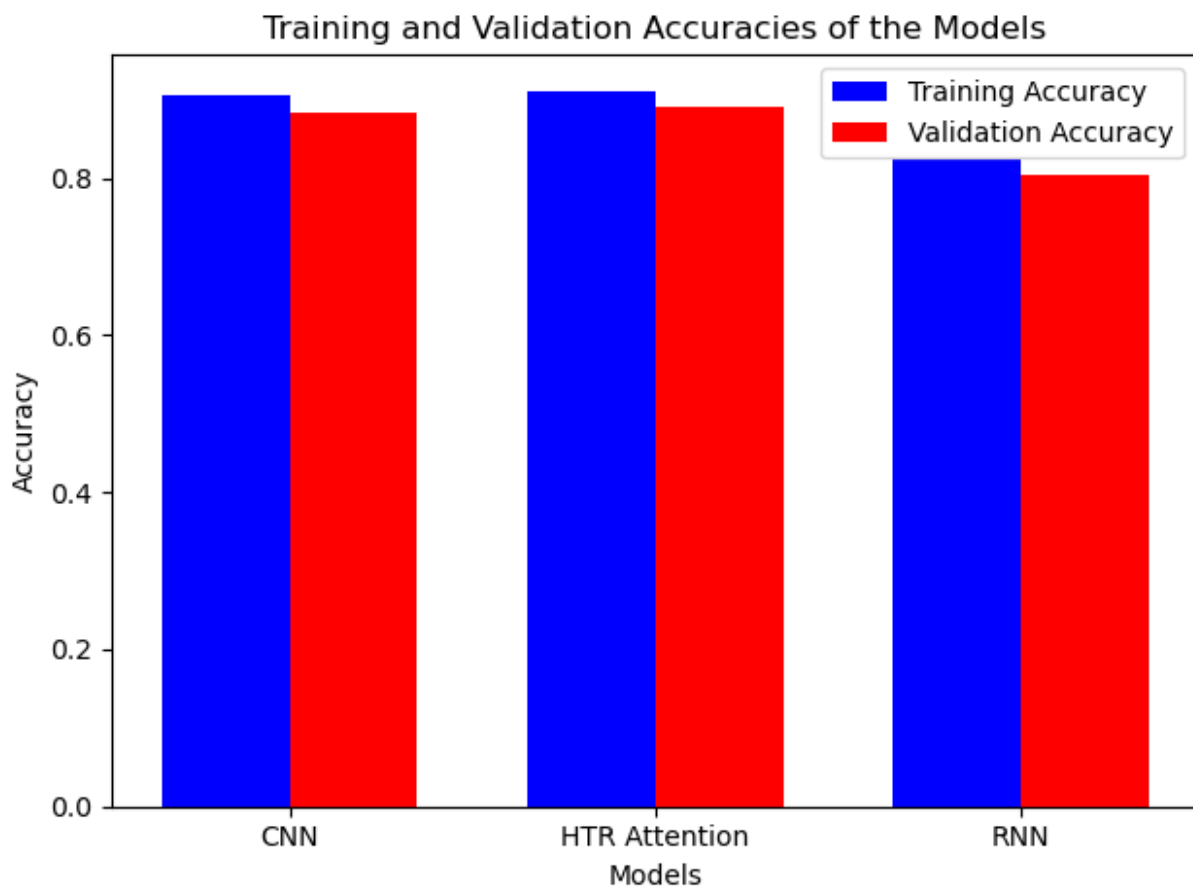
In our pursuit of "Hindko handwritten character recognition utilizing deep neural networks," we have obtained substantial results that underline the effectiveness of our models. The (Table 1) shows accuracy scores and the performance of different deep neural network architectures:

*Table 2: Training and Testing Accuracies of the models.*

| Training and Testing Accuracy of the Models |               |                   |                  |
|---------------------------------------------|---------------|-------------------|------------------|
| Sr.No                                       | Model Name    | Training Accuracy | Testing Accuracy |
| 1                                           | CNN           | 90.57%            | 88.36%           |
| 2                                           | HTR-Attention | 91.13%            | 88.97%           |
| 3                                           | RNN           | 82.49%            | 80.45%           |

These outcomes, as depicted in the bar chart (Figure 2) below, vividly illustrate the capability of our models in recognizing handwritten Hindko characters. Exhibiting the highest level of accuracy throughout both the training and testing stages, the HTR Attention Model demonstrates exceptional performance. This is the case throughout the entire process. The fact

that this is the case demonstrates that Hindko character recognition has the potential to be successful in real-world situations. As a result of these accomplishments, the significance of our research in enhancing Hindko language processing and digital character recognition is demonstrated. In this day and age of digital technology, this helps to ensure that the Hindko language is preserved and can be accessed through the internet.



**Figure 3:** Training vs Validation Accuracies of the Models

At each point in time during testing and training, this bar chart displays the accuracy scores of three different neural networks: CNN, HTR Attention Model, and RNN.

#### **4.4 Class wise performance of the Models:**

The HTR-Attention model's class-wise accuracy scores tell us a lot about how well it works with each type of Hindko character. The classes that are mostly correct and the classes that aren't will be looked at:

*Table 3: Classes with Good Accuracy*

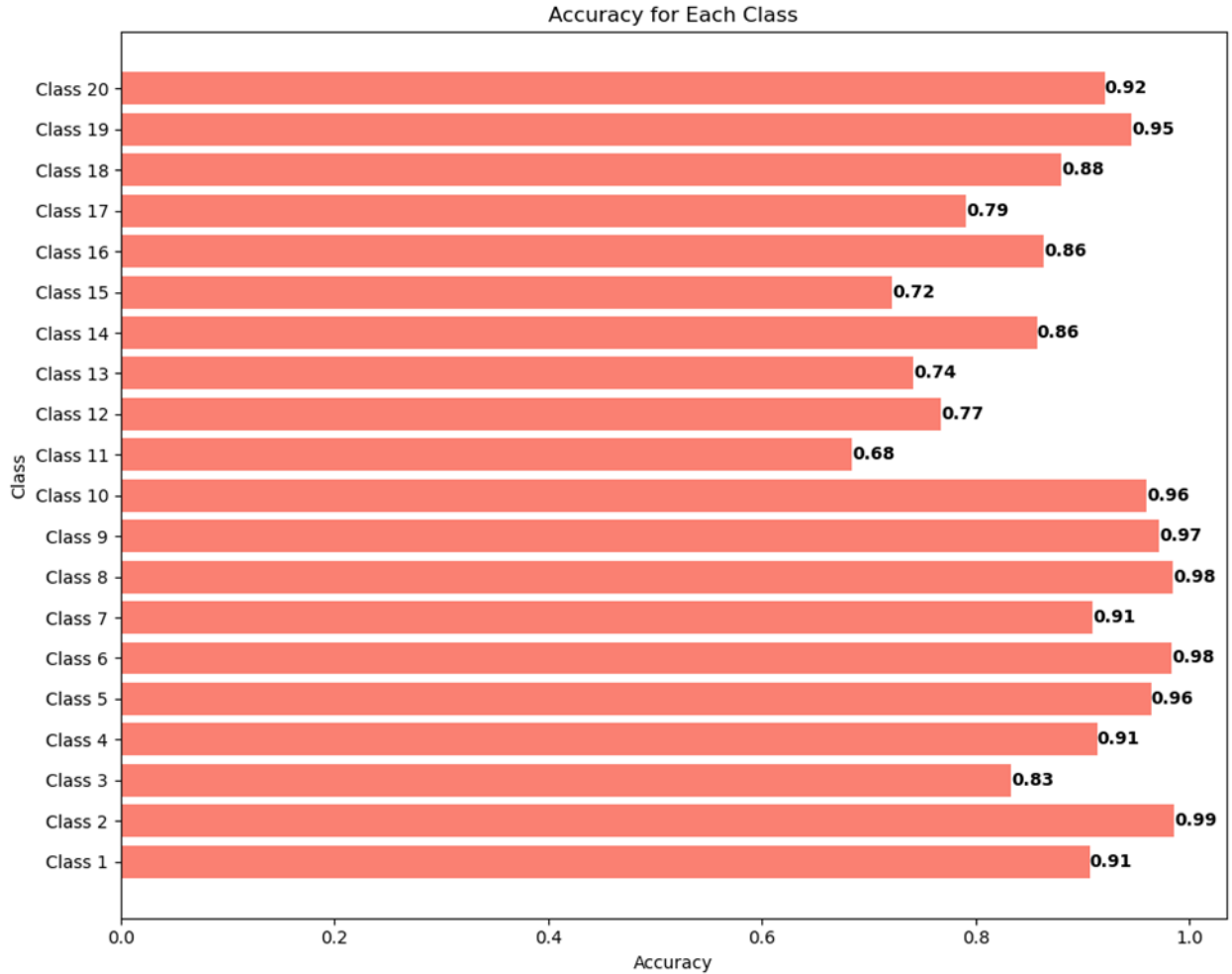
| <b>Classes with Good Accuracy</b> |              |                 |
|-----------------------------------|--------------|-----------------|
| <b>Sr.No</b>                      | <b>Class</b> | <b>Accuracy</b> |
| 1                                 | Class 2      | 98.56%          |
| 2                                 | Class 6      | 98.32%          |
| 3                                 | Class 8      | 98.69%          |
| 4                                 | Class 9      | 97.66%          |
| 5                                 | Class 5      | 95.29%          |

The above classes are correctly classified by the model. Accuracy of above classes is comparatively higher than following classes.

*Table 4: Classes with Lower Accuracy*

| <b>Classes with Lower Accuracy</b> |              |                 |
|------------------------------------|--------------|-----------------|
| <b>Sr.No</b>                       | <b>Class</b> | <b>Accuracy</b> |
| 1                                  | Class 11     | 73.95%          |
| 2                                  | Class 13     | 74.08%          |
| 3                                  | Class 15     | 75.42%          |
| 4                                  | Class 12     | 80.02%          |
| 5                                  | Class 17     | 80.93%          |

The above classes are not correctly classified by the model and show lower accuracy than classes that discussed before. For displaying higher and lower accuracies of the classes we have created a bar plot in which every bar represents the accuracy of the class. Hight of the bar represents the accuracy of the class.



**Figure 4:** Accuracy of Each Class

By above chart a clear visual representation of the performance of the model across different classes is displayed. This visual representation makes it easier to identify the areas where model is performing well and where model is not performing well and need to improve that classes.

#### 4.5 Precision, Recall and F1-Score:

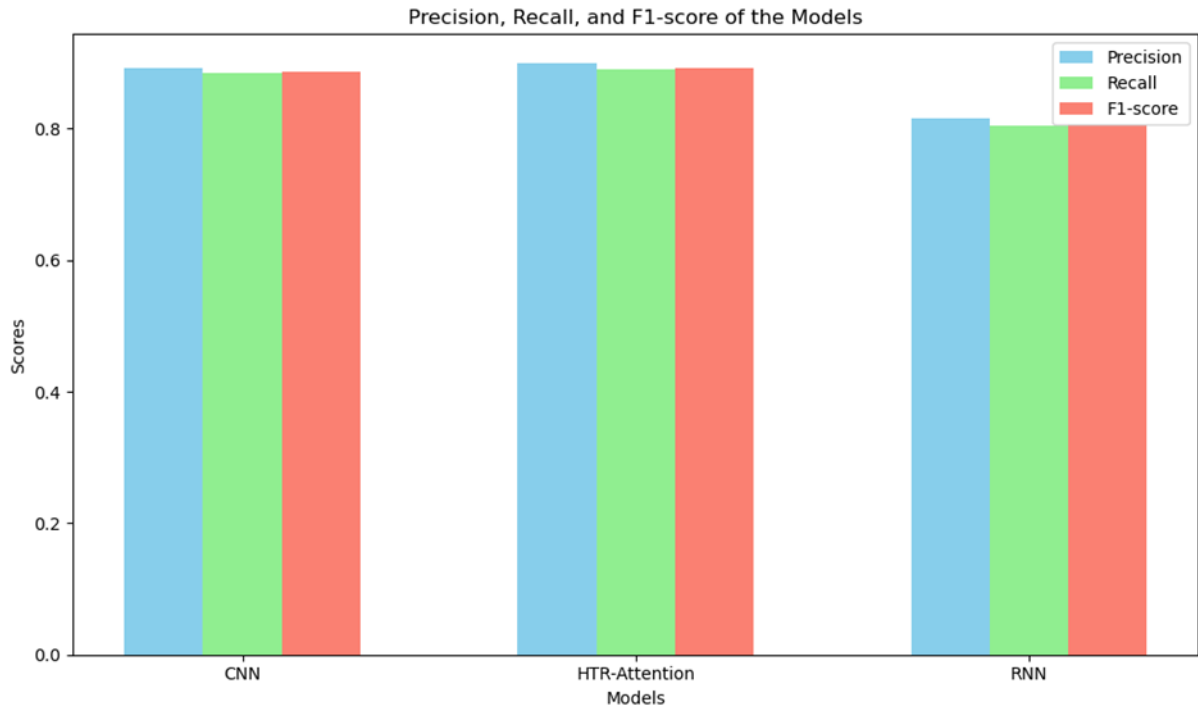
The evaluation metrics like Precision, Recall and F1-Score provide a comprehensive evaluation of how well the CNN, HTR Attention Model and RNN performed in recognizing Hindko handwritten text recognition. Now we conduct an in-depth analysis of these values, in

context of the previous discussion and taking into consideration the bar charts that were created:

*Table 5: Precision, Recall and F1-Score of the Models*

| <b>Precision Recall and F1-Score of the Models</b> |               |                  |               |                 |
|----------------------------------------------------|---------------|------------------|---------------|-----------------|
| <b>Sr.No</b>                                       | <b>Model</b>  | <b>Precision</b> | <b>Recall</b> | <b>F1-Score</b> |
| 1                                                  | CNN           | 0.8917           | 0.8836        | 0.8858          |
| 2                                                  | HTR-Attention | 0.8985           | 0.8897        | 0.8925          |
| 3                                                  | RNN           | 0.8160           | 0.8045        | 0.8064          |

High precision, recall, and an F1-score for its performance show that the CNN model has a well-balanced performance. This suggests that the model can correctly group Hindko characters while also cutting down on the number of false positives and false negatives by a large amount. The HTR Attention Model gets a great F1-score because its precision and recall scores are a little higher than those of the CNN model. Based on this, it looks like the attention mechanism in the HTR model helps recognize Hindko fonts more accurately. There is still a good F1-score for the RNN model, even though it is not as accurate or recalls stories as well as the other models. It's possible that this model isn't as good as CNN and HTR, but it can still be used to recognize Thai characters. There is a chance that the bar chart will have lower scores for accuracy and recall than the other models.



**Figure 5:** Precision, Recall, and F1-score of the Models

Additionally, the CNN model does well, striking a good balance between recall and accuracy. However, it is a good option, though not quite as good as the RNN model. To get a better idea of how the models were alike and different, the bar charts demonstrate these variations visually.

#### 4.6 Mean Square Error:

The performance of CNN model, HTR-Attention Model, and the RNN model can be understood well from the value of Mean square error. The Mean Squared Error (MSE) values tell you a lot about the CNN Model, the HTR-Attention Model, and the RNN Model. These values show that how model truly fit on the training data and how the model predict the future values of the data. We need to look at this MSE data right away:

*Table 2 Mean Square Error of the Models*

| Mean Square Error of the Models |               |                   |
|---------------------------------|---------------|-------------------|
| Sr. No                          | Model         | Mean Square Error |
| 1                               | CNN           | 5.63              |
| 2                               | HTR-Attention | 5.06              |

|   |     |       |
|---|-----|-------|
| 3 | RNN | 11.32 |
|---|-----|-------|

Mean Square Error (MSE) of CNN is minimal which show a notable degree of precision in its forecasts. These minimal values of the CNN shows that model successfully reduces the square deviations between predicted and actual values which show satisfactory performance of the model on testing data. The HTR-Attention model shows slightly lower mean square error than CNN model which show that predictive accuracy of HTR-attention model is higher than CNN model. The results show that adding the attention mechanism to the HTR model helps make predictions more accurate. The RNN model does better than both the CNN and HTR-Attention models in terms of Mean Squared Error (MSE). Based on the observation above, it seems that the RNN model doesn't match the data as well as the other models, which causes predictions to be less accurate.

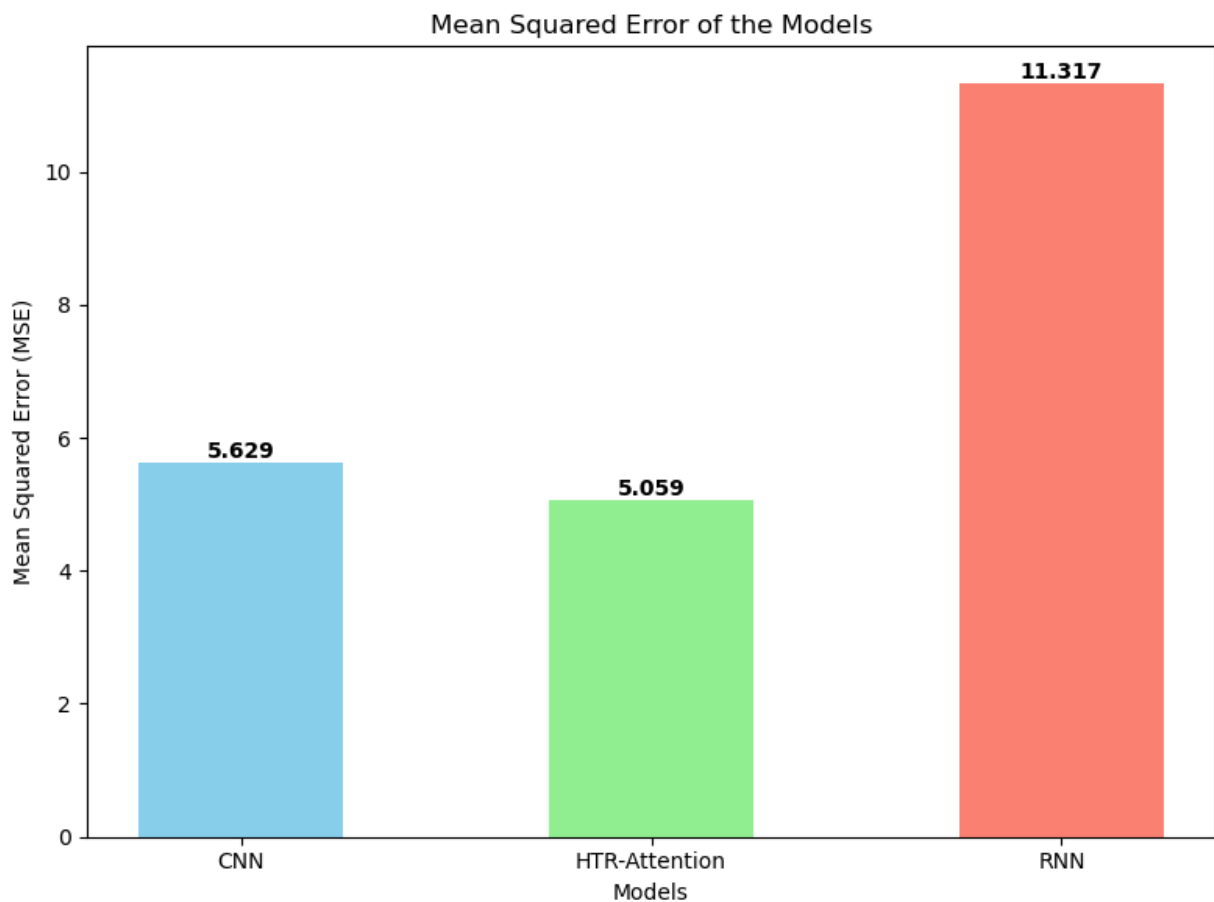


Figure 6: Mean Square Error of the Models

## 4.7 Confusion Matrix:

In the confusion matrix, we can see how well the model identified each group of Hindko handwritten characters. Each row in the matrix represents the actual class, while each column represents the projected class.

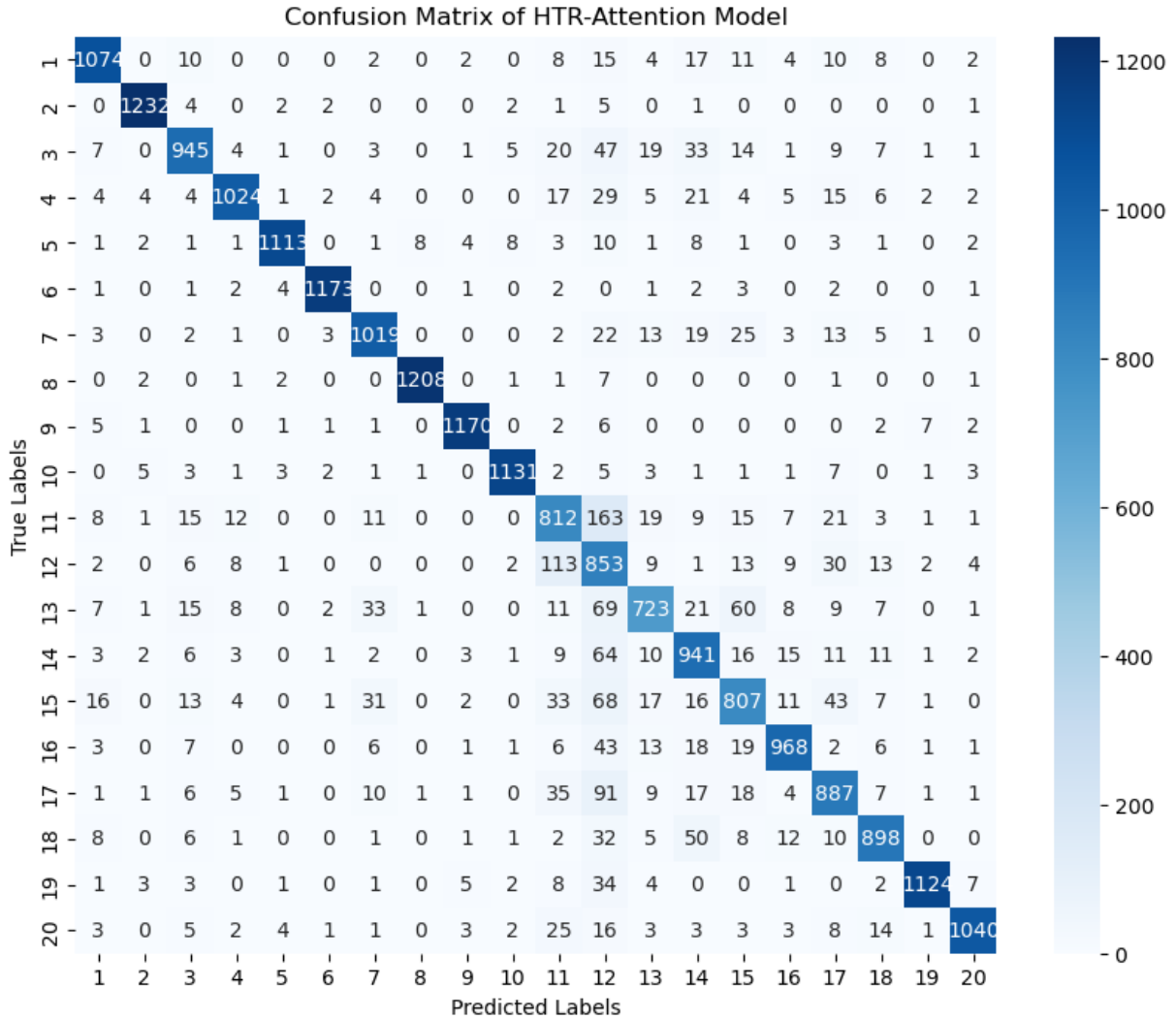


Figure 7: Confusion Matrix of HTR-Attention Model

Here's an analysis of the confusion matrix:

The number of correctly classified cases for each class is represented by the diagonal elements, which run from top left to bottom right. These elements are known as true positives. These numbers should ideally be high, signifying precise forecasts. Off-diagonal elements represent misclassifications. For example, the element at (Row 1, Column 3) indicates that there were 12

instances of characters from class 1 that were misclassified as class 3. Let's analyze specific points:

Class 2 appears to have very few misclassifications, as indicated by the low values in its row (Row 2).

Class 11 and class 13 seem to have relatively higher misclassification rates, as evidenced by non-zero values in multiple columns corresponding to these classes. To evaluate the model's overall performance, one can use metrics such as recall, accuracy, precision, and F1-score. The F1-score takes into consideration the true positives, false negatives, and overall accuracy for each class.

## **Chapter 5**

### **Conclusion and Future Work**

## **5.0 Conclusion and Future Work:**

### **5.1 Conclusion:**

To sum up, this work represents a huge step forward in the digitization of the Hindko language and in the preservation of cultural heritage through its use of deep neural networks to identify Hindko handwritten characters. Hindko is a popular regional language with more than eight million native speakers, but digital representation of Hindko still faces problems due to lack of mature Optical Character Recognition (OCR) system. A lot of testing and evaluation has shown that our research on using deep learning models to identify Hindko characters has led to positive results. An impressive accuracy of 88.36% and 88.97% is achieved by Convolution Neural Network (CNN) and HTR-Attention model respectively. This achievement will help the language translation technologies in recognition of the text.

Various steps including gathering, balancing, preprocessing and choosing models for large set of data were included in research methodology. By making sure the quality and representativeness of the dataset, this strong method made it easier to make recognition models that could be relied on. We also closely looked at the number of performance metrics, including precision, recall, F1-score, Mean Squared Error (MSE), and confusion metrics to get a full picture of how well our models work. It was shown that the HTR Attention Model was better at recalling information and being accurate, which meant it could correctly identify Hindko characters. Since Hindko OCR development is still in its early stages, this study sets the stage for future work that will include expanding the dataset, integrating it with efforts to protect cultural heritage, and creating a full OCR system for Hindko. This study adds to the field of Hindko character recognition and has wider implications for reading handwritten text and keeping languages alive. In the areas of regional languages and preserving cultural heritage, there are new ways to do research and make technological advances.

### **5.2 Future Work**

In this area, one thing that will be done in the future is the creation of a standard dataset for Hindko OCR development. We already have this dataset available. This dataset is supposed to be the building block for a full Hindko optical character recognition system. As we have created first Hindko language dataset now more research and development can be done in this area. Our work schedule for the next few weeks is shown below:

- **Dataset Expansion:** Our first future work schedule is dataset expansion. For the expansion of dataset, it is recommended to collect more comprehensive dataset that consist of wide range of sources, writing styles, regions, and demographics. This will boost the model's ability to adapt to diverse writing styles.
- **Hyperparameter Tuning:** In future Hyperparameter tuning techniques can be used to identify configurations that effectively improve both accuracy and convergence.
- **Advanced Data Augmentation:** Advanced data augmentation techniques can be used for better handling of writing style variations. For example, a data augmentation technique like GANs allows you to increase the dataset size without sacrificing quality.
- **Ensemble Learning:** Ensemble learning techniques can also be used in future. Investigate techniques such as model stacking, bagging, and boosting to strengthen the system and increase recognition accuracy by combining numerous deep learning models.
- **Transfer Learning:** if Transfer learning from previously trained models is applied to larger dataset of Hindko than this could potentially lead to significant improvements in Hindko character recognition accuracy and convergence.

## References

1. Smith J, Johnson K. Fine-tuning vision encoder–decoder transformers for handwriting text recognition on historical documents. In: International Conference on Document Analysis and Recognition; 2023 8 19; 253-268.
2. Barrere K, Soullard Y, Lemaitre A, Coüasnon B. A light transformer-based architecture for handwritten text recognition. In: International Workshop on Document Analysis Systems; 2022 5 18; 275-290.
3. Belhe S, P. C. Hindi handwritten word recognition using HMM and symbol tree. In: Proceeding of the workshop on Document Analysis and Recognition; 2012 10 16. 9-14.
4. Boymatova M. HANDWRITTEN TEXT IMAGE RECOGNITION ALGORITHMS. In: The XXII International Scientific and Practical Conference «Modern scientific space and learning in special conditions; 05–07, 311. 382.
5. Chua LO. CNN: A vision of complexity. *International Journal of Bifurcation and Chaos*. 7(10):2219-2425.
6. Coquenat D, Chatelain C, Paquet T. End-to-end handwritten paragraph text recognition using a vertical attention network. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. 45(1):508-24.
7. Deore SP. Devanagari handwritten character recognition using fine-tuned deep convolutional neural network on trivial dataset. *Sādhanā*. 2020;45:1-13.
8. Dutta K, K. P. Towards accurate handwritten word recognition for Hindi and Bangla. In: Computer Vision, Pattern Recognition, Image Processing, and Graphics: 6th National Conference, NCVPRIPG 2017, 16-19, 20.
9. El-Awadly EM, Ebada AI, Al-Zoghby AM. Arabic Handwritten Text Recognition Systems and Challenges and Opportunities. *The Egyptian Journal of Language Engineering*;10(2):84-103.

10. Hodel TM, Schoch DS, Schneider C, Purcell J. General models for handwritten text recognition: feasibility and state-of-the-art. German kurrent as an example. *Journal of open humanities data*. 7(13):1-0.
11. Hu S, Wang Q, Huang K, Wen M, Coenen F. Retrieval-based language model adaptation for handwritten Chinese text recognition. *International Journal on Document Analysis and Recognition (IJДАР)*. ;26(2):109-19.
12. Indian A, B. K. A survey of offline handwritten Hindi character recognition. *In: 2017 3rd International Conference on Advances in Computing, Communication & Automation (ICACCA)(Fall); 2017 9 15*. 1-6.
13. H, J. A database for handwritten text recognition research. *IEEE Transactions on pattern analysis and machine intelligence*. 1994 5;16(5):550-4.
14. Kabakus AT. A novel handwritten Turkish letter recognition model based on convolutional neural network. *Concurrency and Computation: Practice and Experience*. 2021;33(21):6429.
15. Khan S, Nazir S, Khan HU, Hussain A. Pashto characters recognition using multi-class enabled support vector machine.
16. Khan KR. PHND: Pashtu Handwritten Numerals Database and deep learning benchmark. *Plos one*. 2020;15(9):0238423.
17. KO MA. OCR-nets: variants of pre-trained CNN for Urdu handwritten character recognition via transfer learning. *Procedia computer science*. 2020;171:2294-2301.
18. Kobayashi Y, Mimuro S, Suzuki SN, Iijima Y, Okada A. Basic research on a handwritten note image recognition system that combines two OCRs. *Procedia Computer Science*. 2021 1, 1;192:2596-605.
19. Kumari L, Singh S, Rathore VV, Sharma A. A comprehensive handwritten paragraph text recognition system: Lexiconnet. *In: International Conference on Document Analysis and Recognition; 2023 15; 226-241*.

20. Lukasik EC. Recognition of handwritten Latin characters with diacritics using CNN. Bulletin of the Polish Academy of Sciences: Technical Sciences.
21. G M, G V. KiAn architecture for handwritten text recognition systems. *International Journal on Document Analysis and Recognition*. 1999; 2:37-44.
22. Meng BG, Ghena B. Research on text recognition methods based on artificial intelligence and machine learning.. 2023.
23. Modi N. Text line detection and segmentation in Handwritten Gurumukhi Scripts. *International Journal of Advanced Research in Computer Science and Software Engineering*. 2013;3(5).
24. Nahar KM, Alsmadi I, Al Mamlook RE, Nasayreh A, Gharaibeh H, Almuflih AS, Alasim F. Recognition of Arabic Air-Written Letters: Machine Learning, Convolutional Neural Networks, and Optical Character Recognition (OCR) Techniques. *Sensors*. 2023 Nov 28;23(23).
25. Ngo TT, Nguyen HT, Ly NT, Nakagawa M. Recurrent neural network transducer for Japanese and Chinese offline handwritten text recognition. In: International Conference on Document Analysis and Recognition; 2021 2; 364-376.
26. Nguyen HT, Nguyen CT, Oka H, Ishioka T, Nakagawa M. Handwriting Recognition and Automatic Scoring for Descriptive Answers in Japanese Language Tests. In: International Conference on Frontiers in Handwriting Recognition; 2022 25; 274-284.
27. Nurseitov D, Bekbolat K. Classification of handwritten names of cities and handwritten text recognition using various deep learning models.:2102.04816. 2021 9.
28. Parres D, Rodriguez P. Fine-tuning vision encoder–decoder transformers for handwriting text recognition on historical documents. In: International Conference on Document Analysis and Recognition; 2023 19; 253-268.
29. Sanchez JA, Valveny R. Competition on handwritten text recognition on the READ dataset. In: 2016 15th International Conference on Frontiers in Handwriting Recognition (ICFHR); 2016 23. 630-635.

30. Sanjrani AA. Handwritten optical character recognition system for Sindhi numerals. In: 2016 International Conference on Computing, Electronic and Electrical Engineering (ICE Cube); 2016 4, 262-265
31. Shabir M, Islam N, Jan Z, Khan I. Transformation Invariant Pashto Handwritten Text Classification and Prediction. *Journal of Circuits, Systems and Computers*. 2023 1, 30;32(02):2350020.
32. Truong QV. Vietnamese handwritten character recognition using convolutional neural network. *IAES International Journal of Artificial Intelligence*. 2020;9(2):276.
33. Wang Y, Xiao W, Li S. Offline handwritten text recognition using deep learning: A review. In: *Journal of Physics: Conference Series*; 2021 1; 1848, 1. 012015.
34. Wang ZR, Du J, Wang JM. Writer-aware CNN for parsimonious HMM-based offline handwritten Chinese text recognition. *Pattern Recognition*. 2020 1;100:107102.
35. Yadav M, Purwar RK, Mittal M. Handwritten Hindi character recognition: a review. *IET Image Processing*. 2018;12(11):1919-33.
36. Yu H, Chen J, Li B, Ma J, Guan M, Xu X, Wang X, Qu S, Xue X. Benchmarking Chinese text recognition: Datasets, baselines, and an empirical study. 2112.15093. 2021 30.
37. Deore SP. Devanagari handwritten character recognition using fine-tuned deep convolutional neural network on trivial dataset. *Sādhanā*. 2020;45:1-13.
38. Dutta KP. Towards accurate handwritten word recognition for Hindi and Bangla. In: *Computer Vision, Pattern Recognition, Image Processing, and Graphics: 6th National Conference, NCVPRIPG 2017*, 16-19, 20.
39. Indian ABK. A survey of offline handwritten Hindi character recognition. In: 2017 3rd International Conference on Advances in Computing, Communication & Automation (ICACCA)(Fall); 2017 15. 1-6.
40. H JJ. A database for handwritten text recognition research. *IEEE Transactions on pattern analysis and machine intelligence*. 1994;16(5):550-4.

41. Kabakus AT. A novel handwritten Turkish letter recognition model based on convolutional neural network. *Concurrency and Computation: Practice and Experience*. 2021;33(21):6429.
42. Khan KR. PHND: Pashtu Handwritten Numerals Database and deep learning benchmark. *Plos one*. 2020;15(9):0238423.
43. KO MA. OCR-nets: variants of pre-trained CNN for Urdu handwritten character recognition via transfer learning. *Procedia computer science*. 2020;171:2294-2301.
44. Lukasik EC. Recognition of handwritten Latin characters with diacritics using CNN. *Bulletin of the Polish Academy of Sciences: Technical Sciences*.
45. G MG. ViAn architecture for handwritten text recognition systems. *International Journal on Document Analysis and Recognition*. 1999; 2:37-44.
46. Modi N. Text line detection and segmentation in Handwritten Gurumukhi Scripts. *International Journal of Advanced Research in Computer Science and Software Engineering*. 2013;3(5).
47. Nurseitov DBK. Classification of handwritten names of cities and handwritten text recognition using various deep learning models. 2021 9 2102.04816..
48. Parres DR. Fine-tuning vision encoder–decoder transformers for handwriting text recognition on historical documents. In: *International Conference on Document Analysis and Recognition*; 2023 19; 253-268.
49. Sanchez JAV. Competition on handwritten text recognition on the READ dataset. In: *2016 15th International Conference on Frontiers in Handwriting Recognition (ICFHR)*; 2016 10, 23. 630-635.
50. Sanjrani AA. Handwritten optical character recognition system for Sindhi numerals. In: *2016 International Conference on Computing, Electronic and Electrical Engineering (ICE Cube)*; 2016 262-266.
51. Truong QV. Vietnamese handwritten character recognition using convolutional neural network. *IAES International Journal of Artificial Intelligence*. 2020;9(2):276.

52. Yadav M, Purwar RK, Mittal M. Handwritten Hindi character recognition: a review. IET Image Processing. 2018 11;12(11):1919-33.